A Probabilistic Age Model for Coral Geochemical Records Integrating X-Ray Banding, Climate Constraints, and User-Defined Tie Points

1 Motivation and Overview

Coral archives provide high-resolution records of past climate variability, but converting spatially sampled coral geochemistry into calendar time requires an age model that accounts for variable growth rates, uncertainty in band identification, and incomplete or overlapping sampling paths. We develop a probabilistic age model that integrates structural information from X-ray radiography, seasonal information from climate-driven geochemical variability, and optional user-defined tie points corresponding to known events. The model produces a posterior distribution over the age of every coral sample, allowing uncertainty to be rigorously propagated into downstream analyses.

2 Observed Data

Let samples be indexed by (p, i), where $p = 1, \ldots, P$ denotes sampling transects (parallel monolithic paths) and $i = 1, \ldots, N_p$ indexes samples within transect p, ordered from oldest to youngest.

Observed quantities include:

- $y_{p,i}$: measured coral geochemistry (e.g., δ^{18}O)
- $\text{age2}_{p,i} \in \{1, \ldots, K\}$: annual band index (1 = oldest)
- $\text{age1}_{p,i} \in \{0, 1\}$: indicator for start of a dark X-ray band

Additional information:

- All samples have constant physical width (1.5 mm)
- Sampling paths may overlap in time or leave gaps
- The coral was alive at collection; youngest samples are near April 2007
- A modeled monthly climatology μ_m of coral δ^{18}O derived from observed climate

3 Latent Age Model

3.1 Calendar Time Assignment

Each sample (p, i) has an unknown calendar time $t_{p,i} \in \mathbb{R}$ (decimal years). Within each transect, time must increase monotonically:

$$t_{p,1} < t_{p,2} < \cdots < t_{p,N_p}.$$

Annual bands define year-level segmentation:

$$k(p, i) = \text{age2}_{p,i} - 1 \in \{0, \ldots, K - 1\}.$$

3.2 Year Durations (Interannual Growth Variability)

Each year k has a duration d_k:

$$d_k \sim \mathcal{N}(1, \sigma^2_{\text{year}}), \quad d_k > 0.$$

This prior regularizes year length while allowing biologically plausible variability.

3.3 Within-Year Warping (Seasonal Growth Structure)

Within each annual band, samples are indexed by normalized position $s \in [0, 1)$. The mapping from s to fractional year time is modeled as a monotone piecewise-linear function:

$$g_k(s) = \sum_{j=1}^{M} \Delta_{k,j} \, \mathbb{I}(s \in I_j),$$

with increments constrained by:

$$(\Delta_{k,1}, \ldots, \Delta_{k,M}) \sim \text{Dirichlet}(\alpha, \ldots, \alpha), \quad \sum_{j=1}^{M} \Delta_{k,j} = 1.$$

The Dirichlet prior enforces monotonicity and smoothness while permitting variable seasonal growth.

3.4 Time Construction

Let $B_k = \sum_{\ell < k} d_\ell$ be the start time of year k. The calendar time of sample (p, i) is:

$$t_{p,i} = B_{k(p,i)} + d_{k(p,i)} \cdot g_{k(p,i)}(s_{p,i}).$$

4 Geochemical Observation Model

Observed coral geochemistry is modeled as:

$$y_{p,i} = \mu_{m^*_{p,i}} + A_{k(p,i)} + \varepsilon_{p,i},$$

where:

- $m^*_{p,i}$ is the month implied by $t_{p,i}$ and a phase offset ϕ
- $A_k \sim \mathcal{N}(0, \sigma_A^2)$ is a year-specific anomaly
- $\varepsilon_{p,i} \sim \mathcal{N}(0, \sigma_y^2)$

The phase-adjusted month is defined as:

$$m^*_{p,i} = (\lfloor 12 \cdot \{t_{p,i}\} \rfloor + \phi) \bmod 12 + 1,$$

where $\{t\}$ denotes the fractional part of t.

Year effects A_k are analytically marginalized, yielding a multivariate normal likelihood within each year.

5 X-Ray Band Start Constraint

Samples marked as dark-band starts ($\text{age1}_{p,i} = 1$) are assumed to occur preferentially during a warm-season month m_{warm}. We impose a soft likelihood:

$$\mathcal{L}_{\text{band}} = \prod_{\text{age1}_{p,i}=1} \exp\left(-\frac{d(m^*_{p,i}, m_{\text{warm}})^2}{2\sigma^2_{\text{phase}}} \right),$$

where $d(\cdot,\cdot)$ denotes circular distance on the monthly cycle. This allows uncertainty in band expression and identification.

6 User-Defined Tie Points

Let $j = 1, \ldots, J$ index hand-identified tie points corresponding to known events. Each tie point has:

- observed calendar time $t^\star_j$
- observed coral location (p_j, i_j)
- user-defined uncertainty $\sigma_{t,j}$

We include these as Gaussian constraints:

$$\mathcal{L}_{\text{tie}} = \prod_{j=1}^{J} \mathcal{N}\big(t_{p_j,i_j} \mid t_j^\star,\ \sigma_{t,j}^2\big)\,.$$

7 Multiple Transects with Overlaps and Gaps

Each transect p has its own monotone time mapping $t_p(\cdot)$, but all mappings refer to a common calendar time axis. Transects may overlap or leave gaps.

Let $\mathcal{O}$ denote a set of overlap pairs $((p,i),(q,r))$ believed to represent the same temporal interval. We impose soft time-registration constraints:

$$\mathcal{L}_{\text{overlap-time}} = \prod_{((p,i),(q,r))\in\mathcal{O}} \mathcal{N}\big(t_{p,i} - t_{q,r} \mid 0,\ \sigma_{\Delta t}^2\big)\,.$$

Optionally, geochemical consistency can be enforced:

$$\mathcal{L}_{\text{overlap-geo}} = \prod_{((p,i),(q,r))\in\mathcal{O}} \mathcal{N}\big(y_{p,i} - y_{q,r} \mid 0,\ \sigma_{\Delta y}^2\big)\,.$$

Gaps between transects are treated as unobserved intervals, naturally resulting in increased posterior uncertainty.

8 Absolute Collection-Time Anchor

Because the coral was alive at collection, we impose a soft constraint on the youngest samples:

$$t_{p^*,i^*} \sim \mathcal{N}(t_{\text{2007-04}},\ \sigma_{\text{anchor}}^2),$$

where (p^*, i^*) denotes a sample near the young end, optionally offset by 1–2 samples to reflect spatial uncertainty. A corresponding month constraint enforces proximity to April.

9 Full Posterior Distribution

Let Θ denote all latent parameters:

$$\Theta = \{\mathbf{t}, \mathbf{d}, \mathbf{g}, \phi\}.$$

The full posterior distribution is:

$$\boxed{\begin{aligned} p(\Theta \mid \text{data}) \;\propto\; & \underbrace{\mathcal{L}_{\text{geo}}}_{\text{geochemistry vs climate}} \cdot \underbrace{\mathcal{L}_{\text{band}}}_{\text{X-ray band starts}} \cdot \underbrace{\mathcal{L}_{\text{tie}}}_{\text{user tie points}} \cdot \underbrace{\mathcal{L}_{\text{overlap-time}}}_{\text{transect registration}} \cdot \underbrace{\mathcal{L}_{\text{overlap-geo}}}_{\text{optional geochemical consistency}} \\ & \cdot \underbrace{\mathcal{L}_{\text{anchor}}}_{\text{collection-time constraint}} \cdot \underbrace{\prod_k \mathcal{N}(d_k \mid 1, \sigma^2_{\text{year}})}_{\text{year-length regularization}} \cdot \underbrace{\prod_k \text{Dirichlet}(\boldsymbol{\Delta}_k \mid \alpha)}_{\text{within-year smoothness}} \cdot \underbrace{p(\phi)}_{\text{phase prior}} . \end{aligned}}$$

10 Inference

Inference is performed using particle (ensemble) sampling:

1. Loop over discrete phase offsets $\phi \in \{0, \ldots, 11\}$
2. Sample year durations and warping parameters from their priors
3. Construct calendar times $t_{p,i}$
4. Evaluate the full log-posterior
5. Normalize weights and marginalize

This yields posterior means and credible intervals for all sample ages, as well as the posterior distribution over phase.

11 Conclusion

This framework provides a flexible and physically grounded probabilistic age model for coral archives, capable of integrating structural banding, climate seasonality, manual tie points, and complex sampling geometries. Uncertainty is coherently propagated from all sources into the final age estimates, enabling robust paleoclimate inference.